

Simulating the transient photoelectronic spectrum of azobenzene with surface hopping

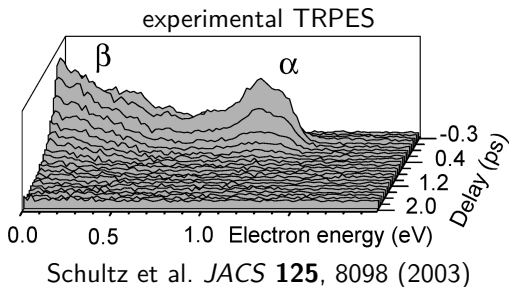
Giovanni Granucci

University of Pisa

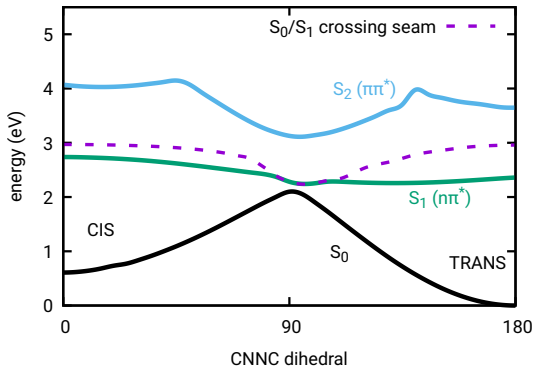
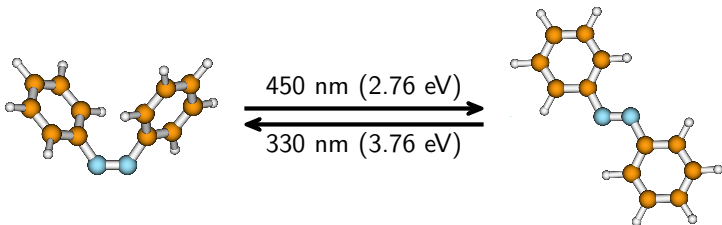
giovanni.granucci@unipi.it

Motivation

- Why another study of *trans*-azobenzene molecule in vacuum?
- Very few gas phase experimental results
- Consensus about the dynamics after $n \rightarrow \pi^*$ excitation
- Dynamics after $\pi \rightarrow \pi^*$ excitation still uncertain



Overview of azobenzene photodynamics



- After $n \rightarrow \pi^*$ excitation:
 $S_1 \rightarrow S_0$ decay through the MECI at $\text{CNNC} \simeq 90^\circ$, with possible isomerization.
 Torsional mechanism, combined with CNN opening (inversion) in hindering environments
- After $\pi \rightarrow \pi^*$ excitation:
 Excess vibrational energy on S_1 may trigger early decay to S_0 , lowering the photoisomerization QY

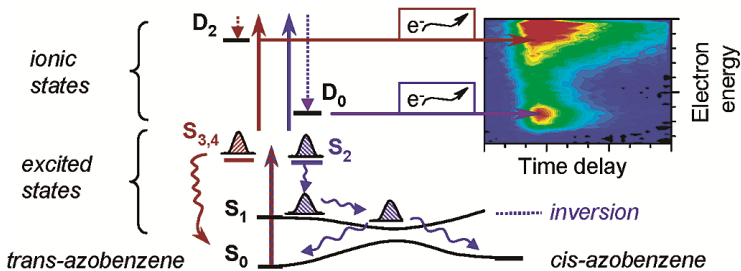
<i>trans</i> $\rightarrow$ <i>cis</i> photoisom. quantum yield		
excitation	expt. (in MeOH)	calc. (in vac.)
$n \rightarrow \pi^*$	0.22–0.31	0.36 ± 0.02
$\pi \rightarrow \pi^*$	0.12–0.16	0.23 ± 0.02

Ladany et al. *Photochem. Photobiol. Sci.* **16**, 1757 (2017)

Cantatore, Granucci, Persico *Comput. Theor. Chem.* **1040**, 126 (2014)

Nenov et al. *JPCL* **9**, 1534 (2018)

Proposed interpretation of the experimental TRPES data



Schultz et al. *JACS* **125**, 8098 (2003)

Method: Dyson orbital

Photoionization cross section (first order, dipole approx.)

$$\sigma_{sf} \propto |\langle \Psi_s^N | \mu_x | \Psi_f^{N-1} \wedge \psi_{k_e} \rangle|^2$$

Ψ_s^N = starting state (molecule) Ψ_f^{N-1} = final state (ion)

ψ_{k_e} = photoelectron wf

Assuming fast photoionization $\implies$ strong orthogonality btw Ψ_s^N and ψ_{k_e}

$$\langle \Psi_s^N | \mu_x | \Psi_f^{N-1} \wedge \psi_{k_e} \rangle = \langle \psi_{sf}^d | \mu_x | \psi_{k_e} \rangle$$

$$\psi_{sf}^d = \sum_j \langle \Psi_f^{N-1} | \hat{a}_j | \Psi_s^N \rangle \psi_j \quad \text{Dyson orbital}$$

Disregarding the wf of the ejected electron

$$\sigma_{sf}(\mathbf{R}) \approx C \langle \psi_{sf}^d(\mathbf{R}) | \psi_{sf}^d(\mathbf{R}) \rangle$$

$\mathbf{R}$ = nuclear coordinates

Photoelectron spectra

Photoelectron signal from N_t classical trajectories $\mathbf{R}_j(t)$

$$S(t, \epsilon) = \sum_{j=1}^{N_t} \frac{1}{N_t} \sum_f \sigma_{sf}(\mathbf{R}_j(t)) \times \delta(\hbar\omega - \Delta E_{sf}(\mathbf{R}_j(t)) - \epsilon)$$

$\hbar\omega$ = probe energy

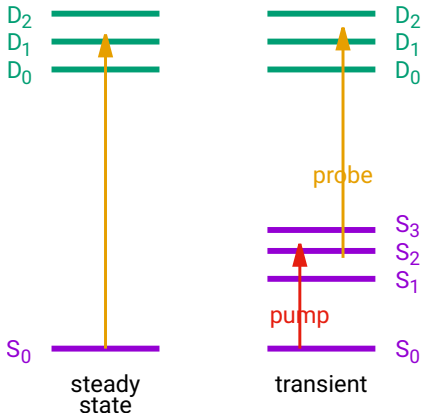
$\epsilon = e^-$ kinetic energy

$\epsilon = \hbar\omega - (E_f - E_s)$

consistently with

fast photoion. approx.

$\mathbf{R}(t)$, $\mathbf{P}(t)$ constant



Arbelo-Gonzalez, Crespo-Otero,
Barbatti, *JCTC* 12, 5037 (2016)

Spin multiplets

Evaluation of the matrix elements $\langle S, M_s | \hat{a}_i | S' M'_s \rangle$

The essential point:

$$\begin{pmatrix} \hat{a}_{i\beta} \\ -\hat{a}_{i\alpha} \end{pmatrix} = \begin{pmatrix} \hat{\mathcal{T}}_{1/2} \\ \hat{\mathcal{T}}_{-1/2} \end{pmatrix}$$

where $\hat{\mathcal{T}}$ is a rank 1/2 irreducible spherical tensor operator.

The Wigner-Eckart theorem can be applied.

For example

$$\langle D^{(1/2)} | \hat{a}_{i\alpha} | T^{(1)} \rangle = \sqrt{2} \langle D^{(1/2)} | \hat{a}_{i\beta} | T^{(0)} \rangle$$

$$\langle D^{(-1/2)} | \hat{a}_{i\beta} | T^{(-1)} \rangle = \sqrt{2} \langle D^{(1/2)} | \hat{a}_{i\beta} | T^{(0)} \rangle$$

Setup for azobenzene

Semiempirical AM1 reparameterized – FOMO-CI approach

Molecule:

Active space of 13 MOs and 14 electrons – 94 determinants

Cation:

Share the same set of semiempirical parameter of the molecule

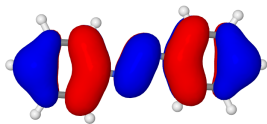
Has its own set of FOMOs

Active space of 13 MOs and 13 electrons – 472 determinants

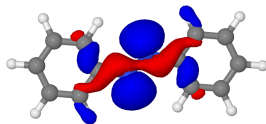
The CI space of the cation is not closed with respect to $\hat{S}^2$. However, doublet states D_0 – D_3 are well described

The zero of the energy of the ion is set so as to reproduce the experimental ionization energy

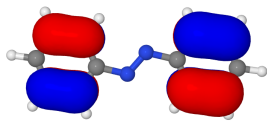
Dyson orbitals



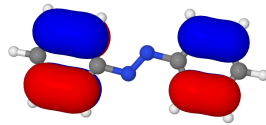
D_0



D_1



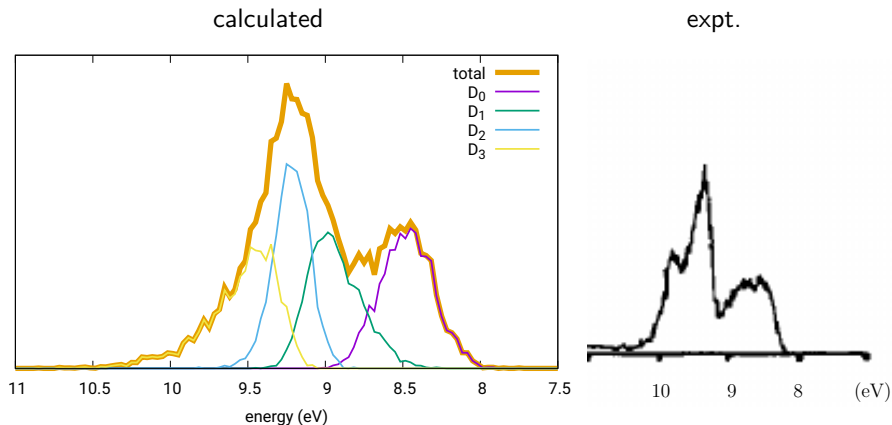
D_2



D_3

Dyson orbitals of azobenzene, evaluated at the S_0 minimum, show very good agreement with *ab initio* results of Carlini et al., *JCP* **158**, 054201 (2023)

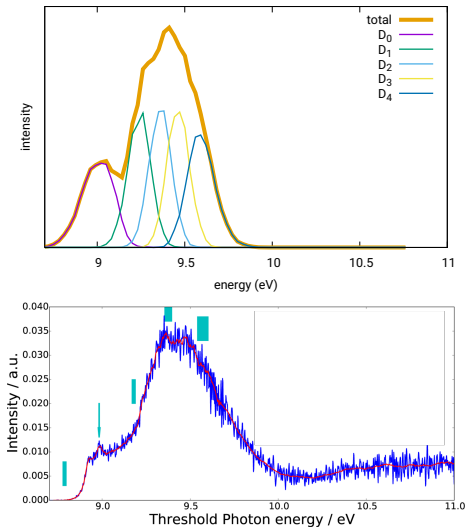
Azobenzene photoelectronic spectrum



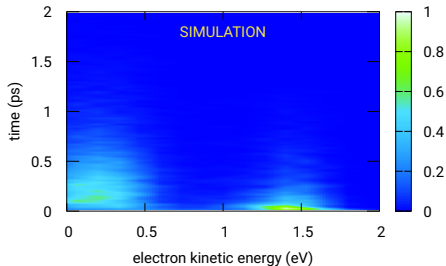
Kobayashi, Yokota, Nagakura, *J. El. Spect. Rel. Phenom.* **6**, 167 (1975)

Carlini et al., *JCP* **158**, 054201 (2023)

Benzophenone photoelectron spectrum

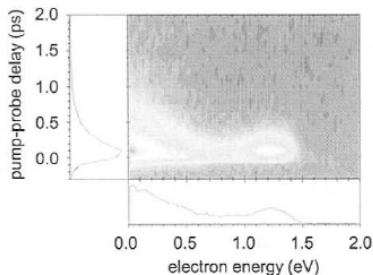


Azobenzene TRPES



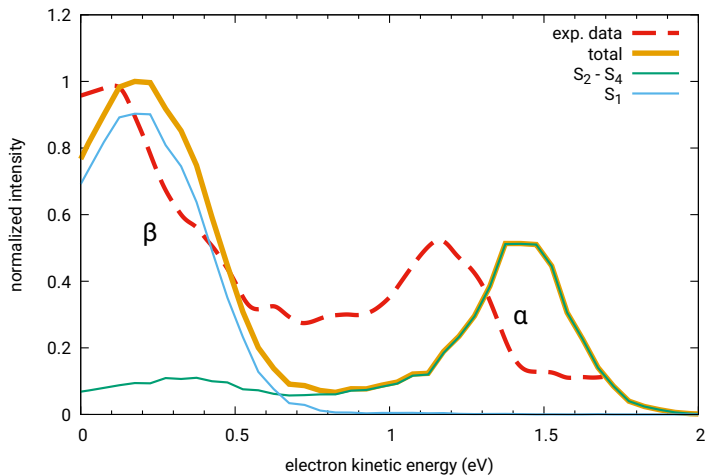
FSSH-ODC trajectories

pump = 3.76 eV
probe = 6.0 eV
exper. $\Delta t = 100$ fs



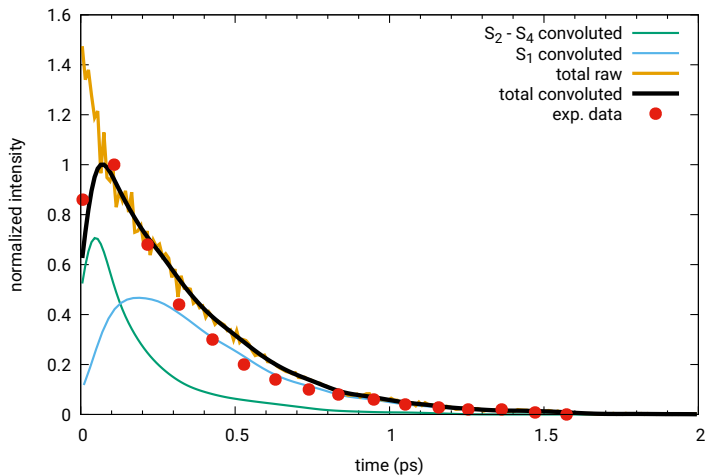
Time-integrated results

Dashed red curve: experimental results



Energy-integrated results, Gaussian convoluted with FWHM = 100 fs

Red filled circles: experimental results



Azobenzene photodynamics with MASH

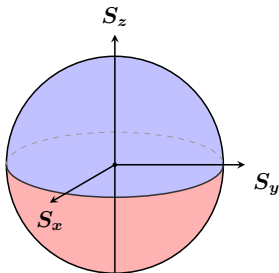
The MASH approach (Richardson and coworkers):

- The original approach is only valid for two states
- Deterministic hops: the active electronic state has the largest population
- Stochastic sampling of the initial electronic coefficients. All the states are initially populated.
- Each trajectory has a different weight depending on the population of the active state at $t = 0$
- Less sensitive than FSSH to decoherence
- Several attempts to extend MASH to multiple states
- In our multistate “semifocused” variant (SMASH), the electronic states are partitioned in cluster according to their energy. Only the states belonging to the cluster of the active state are initially populated.

Mannouch, Richardson, *JCP* **158**, 104111 (2023)

Richardson, Lawrence, Mannouch, *Annu. Rev. Phys. Chem.* **76**, 663 (2025)

2-states MASH



$$|\psi_{el}\rangle = c_1|1\rangle + c_2|2\rangle \quad \rho_{ij} = c_i c_j^*$$

$$S_z = \rho_{22} - \rho_{11}$$

$$S_x = 2\text{Re}(\rho_{12})$$

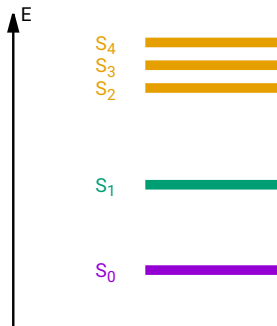
$$S_y = 2\text{Im}(\rho_{12})$$

$$\hbar \dot{\vec{S}} = \vec{B} \times \vec{S} \quad \vec{B} = \begin{pmatrix} 0 \\ 2\hbar \mathbf{d} \cdot \dot{\mathbf{R}} \\ E_2 - E_1 \end{pmatrix}$$

$$\text{traj. weight} \quad w = 2 |S_z(0)|$$

The SMASH approach

1D clustering based on energy
(Donkey algorithm, Kára et al.)



ad-hoc trajectory weights:

$$w = A[n_k \rho_{kk}(0) - 1]^{n_k - 1}$$

$\rho_{kk}(0)$ = initial active state population

n_k = number of states initially populated

A = normalization factor

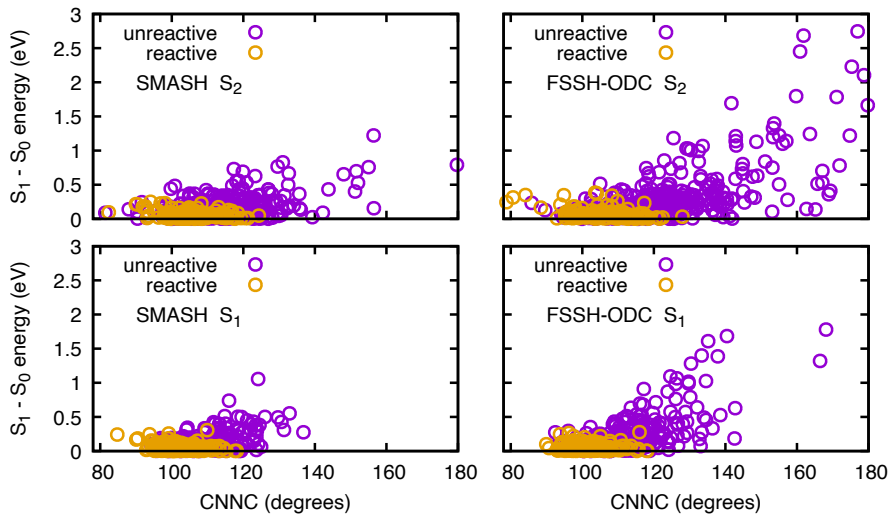
Kara, Acheson, Kirrander, *JCTC* **21**, 5789 (2025)

Botticelli, Accomasso, Granucci *JCP* **163**, 224124 (2025)

lifetimes (in ps)				
excitation	state	FSSH	SMASH	MASH2
$n \rightarrow \pi^*$	S_1	0.38	0.40	0.38
$\pi \rightarrow \pi^*$	S_2	0.12	0.14	—
	S_1	0.35	0.43	—

<i>trans</i> $\rightarrow$ <i>cis</i> quantum yields				
excitation	expt. (in MeOH)	calc. (s.dev.=0.02)		
		FSSH	SMASH	MASH2
$n \rightarrow \pi^*$	0.22–0.31	0.36	0.44	0.41
$\pi \rightarrow \pi^*$	0.12–0.16	0.23	0.28	—

MASH2 $\rightarrow$ standard 2-states MASH



upper panels $\pi \rightarrow \pi^*$ excitation

lower panels $n \rightarrow \pi^*$ excitation

Conclusions

- Simulation results in semi-quantitative agreement with the experimental TRPES
- Validation of FSSH-ODC decay pathway and lifetimes
- Similar results obtained with MASH without decoherence correction
- Quantitative account of quantum yields more difficult

Acknowledgements

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Thank you